

Simplification with 3 variables

Exercise 1

a / bc	00	01	11	10
0	1	1	1	1
1	1	1	1	0

Exercise 2

a / bc	00	01	11	10
0	0	1	1	1
1	1	1	0	0

Exercise 3

a / bc	00	01	11	10
0	1	0	0	1
1	0	1	1	0

Exercise 4

a / bc	00	01	11	10
0	1	0	0	1
1	0	0	1	0

Exercise 5

a / bc	00	01	11	10
0	1	1	0	1
1	1	1	0	0

Exercise 6

a / bc	00	01	11	10
0	0	1	1	0
1	1	1	0	0

Exercise 7

a / bc	00	01	11	10
0	0	1	1	1
1	1	1	0	0

Exercise 8

a / bc	00	01	11	10
0	1	1	1	0
1	0	1	1	1

Exercise 9

a / bc	00	01	11	10
0	1	0	1	0
1	0	1	1	1

Exercise 10

a / bc	00	01	11	10
0	0	0	1	1
1	1	1	1	0

Exercise 11

a / bc	00	01	11	10
0	1	1	1	0
1	1	1	1	1

Exercise 12

a / bc	00	01	11	10
0	1	0	0	1
1	0	1	1	1

Exercise 13

a / bc	00	01	11	10
0	0	0	1	0
1	1	0	0	1

Exercise 14

a / bc	00	01	11	10
0	0	0	0	1
1	0	1	1	1

Exercise 15

a / bc	00	01	11	10
0	1	0	1	1
1	1	1	0	1

Exercise 16

a / bc	00	01	11	10
0	1	1	1	1
1	1	1	0	1

Exercise 1

a / bc	00	01	11	10
0	1	1	1	1
1	1	1	1	0

Solution to Exercise 1

There are 3 prime implicants:

$$\bar{a}$$

$$\bar{b}$$

$$c$$

A minimum cover is made of 3 prime implicants:

$$\bar{a} + \bar{b} + c$$

Exercise 2

a / bc	00	01	11	10
0	0	1	1	1
1	1	1	0	0

Solution to Exercise 2

There are 4 prime implicants:

$$\bar{a}b$$

$$a\bar{b}$$

$$\bar{a}c$$

$$\bar{b}c$$

A minimum cover is made of 3 prime implicants:

$$\bar{a}b + a\bar{b} + \bar{b}c$$

Exercise 3

a / bc	00	01	11	10
0	1	0	0	1
1	0	1	1	0

Solution to Exercise 3

There are 2 prime implicants:

$$a c$$

$$\bar{a} \bar{c}$$

A minimum cover is made of 2 prime implicants:

$$a c + \bar{a} \bar{c}$$

Exercise 4

a / bc	00	01	11	10
0	1	0	0	1
1	0	0	1	0

Solution to Exercise 4

There are 2 prime implicants:

$$\bar{a} \bar{c}$$

$$a b c$$

A minimum cover is made of 2 prime implicants:

$$\bar{a} \bar{c} + a b c$$

Exercise 5

a / bc	00	01	11	10
0	1	1	0	1
1	1	1	0	0

Solution to Exercise 5

There are 2 prime implicants:

$$\bar{b}$$

$$\bar{a} \bar{c}$$

A minimum cover is made of 2 prime implicants:

$$\bar{b} + \bar{a} \bar{c}$$

Exercise 6

a / bc	00	01	11	10
0	0	1	1	0
1	1	1	0	0

Solution to Exercise 6

There are 3 prime implicants:

$$a\bar{b}$$

$$\bar{a}c$$

$$\bar{b}c$$

A minimum cover is made of 2 prime implicants:

$$a\bar{b} + \bar{a}c$$

Exercise 7

a / bc	00	01	11	10
0	0	1	1	1
1	1	1	0	0

Solution to Exercise 7

There are 4 prime implicants:

$$\bar{a}b$$

$$a\bar{b}$$

$$\bar{a}c$$

$$\bar{b}c$$

A minimum cover is made of 3 prime implicants:

$$\bar{a}b + a\bar{b} + \bar{b}c$$

Exercise 8

a / bc	00	01	11	10
0	1	1	1	0
1	0	1	1	1

Solution to Exercise 8

There are 3 prime implicants:

$$\begin{array}{c} c \\ ab \\ \bar{a}\bar{b} \end{array}$$

A minimum cover is made of 3 prime implicants:

$$c + ab + \bar{a}\bar{b}$$

Exercise 9

a / bc	00	01	11	10
0	1	0	1	0
1	0	1	1	1

Solution to Exercise 9

There are 4 prime implicants:

$$\begin{array}{c} ab \\ ac \\ bc \\ \bar{a}\bar{b}\bar{c} \end{array}$$

A minimum cover is made of 4 prime implicants:

$$ab + ac + bc + \bar{a}\bar{b}\bar{c}$$

Exercise 10

a / bc	00	01	11	10
0	0	0	1	1
1	1	1	1	0

Solution to Exercise 10

There are 4 prime implicants:

$$\begin{array}{c} \bar{a}b \\ a\bar{b} \\ ac \\ bc \end{array}$$

A minimum cover is made of 3 prime implicants:

$$\bar{a}b + a\bar{b} + bc$$

Exercise 11

a / bc	00	01	11	10
0	1	1	1	0
1	1	1	1	1

Solution to Exercise 11

There are 3 prime implicants:

$$a$$

$$\bar{b}$$

$$c$$

A minimum cover is made of 3 prime implicants:

$$a + \bar{b} + c$$

Exercise 12

a / bc	00	01	11	10
0	1	0	0	1
1	0	1	1	1

Solution to Exercise 12

There are 4 prime implicants:

$$a b$$

$$a c$$

$$\bar{a} \bar{c}$$

$$b \bar{c}$$

A minimum cover is made of 3 prime implicants:

$$a c + \bar{a} \bar{c} + b \bar{c}$$

Exercise 13

a / bc	00	01	11	10
0	0	0	1	0
1	1	0	0	1

Solution to Exercise 13

There are 2 prime implicants:

$$a\bar{c}$$

$$\bar{a}bc$$

A minimum cover is made of 2 prime implicants:

$$a\bar{c} + \bar{a}bc$$

Exercise 14

a / bc	00	01	11	10
0	0	0	0	1
1	0	1	1	1

Solution to Exercise 14

There are 3 prime implicants:

$$ab$$

$$ac$$

$$b\bar{c}$$

A minimum cover is made of 2 prime implicants:

$$ac + b\bar{c}$$

Exercise 15

a / bc	00	01	11	10
0	1	0	1	1
1	1	1	0	1

Solution to Exercise 15

There are 3 prime implicants:

$$\bar{c}$$

$$\bar{a}b$$

$$a\bar{b}$$

A minimum cover is made of 3 prime implicants:

$$\bar{c} + \bar{a}b + a\bar{b}$$

Exercise 16

a / bc	00	01	11	10
0	1	1	1	1
1	1	1	0	1

Solution to Exercise 16

There are 3 prime implicants:

$$\bar{a}$$

$$\bar{b}$$

$$\bar{c}$$

A minimum cover is made of 3 prime implicants:

$$\bar{a} + \bar{b} + \bar{c}$$